

ICS 3106 – Operations Research and Management Science

Classwork / Assignment – Group E

Unbalanced Transportation and Assignment Problems

Question 1: Unbalanced Transportation Problem

(20 marks)

A Kenyan mobile-network operator ships fibre-optic equipment from four supply depots to three regional hubs. The monthly supply (units), demand (units), and unit shipping costs (KSh $\times 10^3$) are given in Table 1.

Table 1: Shipping costs, supply and demand – Group E

Depot (Source)	Mombasa	Nakuru	Thika	Supply (units)
Nairobi	9	7	11	150
Eldoret	4	8	6	200
Kisumu	10	5	9	130
Meru	6	12	7	170
Demand (units)	220	190	180	650

- (a) Show that this problem is **unbalanced**. Explain the corrective action required and construct the balanced transportation tableau, clearly indicating the dummy destination and its associated costs. (4 marks)
- (b) Using the **Minimum Cell Cost Method**, determine an initial basic feasible solution for the balanced tableau. Show the allocation sequence clearly. (6 marks)
- (c) Check whether the initial solution is **non-degenerate** by verifying the condition $m + n - 1$. (2 marks)
- (d) Apply the **Modified Distribution (MODI) Method** to test and improve the initial solution until optimality is reached. Show all u_i , v_j and k_{ij} computations. (6 marks)
- (e) State the optimal shipping schedule and the **minimum total cost**. Identify which depot retains unused stock. (2 marks)

Question 2:*(20 marks)*

A Kenyan cloud-computing company needs to assign five system administrators to four server-maintenance tasks for the coming week. The estimated effort (person-hours) for each administrator on each task is given in Table 2.

Table 2: Effort estimates (person-hours) – Group E

Administrator	Database	UI Design	Testing	DevOps
Amina	12	9	14	8
Benson	7	11	9	15
Cynthia	10	6	12	11
Dennis	9	14	7	10
Esther	11	8	10	13

- Explain why this assignment problem is **unbalanced**. Show how to balance it and state the costs for any dummy column introduced. *(3 marks)*
- Perform **row reductions** on the balanced table. *(3 marks)*
- Perform **column reductions** on the result of (b). *(3 marks)*
- Apply the **Hungarian Method** to determine the optimal assignment. Show all line tests and any adjustments to the opportunity-cost table. *(8 marks)*
- State the optimal assignment, identify the administrator who is left unassigned, and compute the **minimum total effort**. *(3 marks)*